a. **Prove that** the area under the curve of the probability density function (PDF) f of a normally-distributed random variable, with mean μ and standard deviation σ, equals one.

b. The cumulative distribution function (CDF) F of a continuous random variable X is defined as $F(x) = \int_{-\infty}^{x} f(x)\, dx$, with f being the PDF.

For the PDF f given in part (a), **find** the value of $F(0)$.

Hint: The PDF f of the normally-distributed random variable X, with mean μ and standard deviation σ, is formulated as $f(x) = \frac{\exp[-(x-\mu)^2/2\sigma^2]}{\sqrt{2\pi}\,\sigma}$.

A. a) $A = \int_{-\infty}^{\infty} f(x)dx = \int_{-\infty}^{\infty} \frac{\exp[-(x-\mu)^2/2\sigma^2]}{\sqrt{2\pi}\,\sigma} dx = \frac{1}{\sqrt{2\pi}\,\sigma}\int_{-\infty}^{\infty} e^{-\left(\frac{x-\mu}{\sqrt{2}\,\sigma}\right)^2} dx$

Set: $t = \frac{x-\mu}{\sqrt{2}\,\sigma} \Rightarrow \sqrt{2}\,\sigma dt = dx \Rightarrow A = \frac{1}{\sqrt{2\pi}\,\sigma}\int_{-\infty}^{\infty} e^{-t^2}\sqrt{2}\,\sigma dt = \frac{1}{\sqrt{\pi}}\int_{-\infty}^{\infty} e^{-t^2} dt$ (1)

Let $I = \int_{-\infty}^{\infty} e^{-t^2} dt = \int_{-\infty}^{\infty} e^{-u^2} du \Rightarrow I^2 = \int_{-\infty}^{\infty} e^{-t^2} dt \int_{-\infty}^{\infty} e^{-u^2} du = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-t^2} e^{-u^2} dtdu$

$$\Rightarrow I^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(t^2+u^2)}\, dtdu = \iint_{\mathbb{R}^2} e^{-(t^2+u^2)} dA$$

By using polar coordinates (r, θ)

$$\Rightarrow I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\, drd\theta = \int_0^{2\pi} d\theta \int_0^{\infty} e^{-r^2} r\, dr = [\theta]_0^{2\pi} \lim_{\beta\to\infty}\left[-0.5e^{-r^2}\right]_0^{\beta} = \pi$$

$$\Rightarrow I = \sqrt{I^2} = \sqrt{\pi}\ (2) \overset{(2)\text{ into }(1)}{=} A = 1$$

b)

$$f(0) = 0.5$$